

Roof Lifecycle Cost Analysis

Planned maintenance is projected to save the owner of a typical Culbertson (N. Virginia) school **\$2.31M** over the next 20 years.

That is 64.1% less than the reactive cost, or \$1.01M in today's dollars. It compares PaxSeal planned maintenance with fixing problems only after they leak, using Culbertson (N. Virginia) costs and storm history.

MARKET	PROPERTY	ROOF AREA	SYSTEM	AGE	NEW ROOF TODAY
Culbertson (N. Virginia)	K-12	120,000 sq ft	Mod bit	8 yrs Good	\$1.14M

PAXSEAL NET COST

\$1.3M

\$2.56M spent, less \$1.26M roof life left

REACTIVE NET COST

\$3.61M

\$3.8M spent, less \$187K roof life left

EMERGENCY COSTS CUT

\$1.9M

Unbudgeted repairs and storm damage avoided; \$5.25 back per fee dollar

NEW ROOF PRICE

2× in 15 yrs

\$1.14M today, \$2.8M by year 20

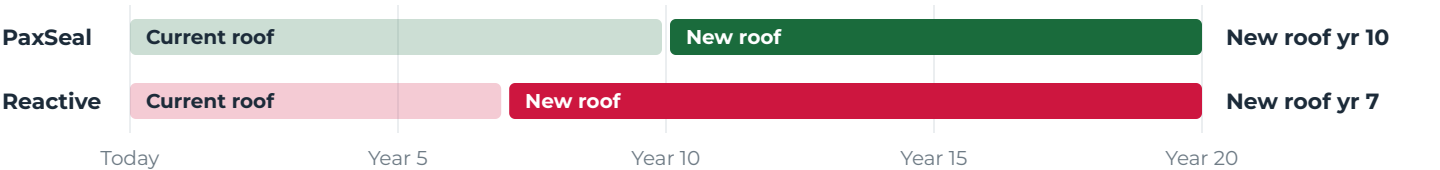
Cumulative roof spending, next 20 years

Nominal dollars. Vertical jumps are full roof replacements.



Roof life

When each scenario needs a new roof.



Where the savings come from

20-year totals

	PAXSEAL	REACTIVE	SAVINGS
PaxSeal program fees	\$441K	—	-\$441K
Repairs paid by owner	\$274K	\$2.15M	+\$1.88M
Expected storm damage	\$57K	\$86K	+\$29K
Roof replacements	\$1.79M	\$1.56M	-\$226K
Credit for roof life left	-\$1.26M	-\$187K	+\$1.07M
Net 20-year cost	\$1.3M	\$3.61M	+\$2.31M

What this means

- 1

A later replacement. Maintenance moves the next new roof from year 7 to year 10, so fewer roofs are bought over the period.
- 2

Smaller repairs. Problems found on a scheduled visit avoid emergency work, which costs about 3× as much. That saves \$1.88M in repairs.
- 3

A predictable budget line. PaxSeal turns unplanned emergency repairs into a fixed annual cost the board can plan for, and keeps leaks from closing classrooms.
- 4

The program pays for itself. Each \$1 of PaxSeal fees returns about \$5.25 in avoided cost.

Local risk profile

Prince William County, VA (51153)

Thunderstorm wind / derecho, snow and ice

Strong wind and winter weather both rated Relatively High (FEMA NRI); tropical remnants ~1 per 12 years. Expected storm damage to this roof over 20 years is **\$57K** if maintained and **\$86K** if neglected. A documented maintenance history also helps when an insurer questions a claim.

Hail	Moderate	
Strong wind	High	
Tornado	Moderate	
Hurricane	Moderate	
Winter weather	High	
Ice storm	Low	

FEMA National Risk Index rating

Key assumptions

Confidence reflects source strength. Values will be refined as research continues.

ASSUMPTION	VALUE	BASIS	CONFIDENCE
Roof life	20 / 15 yrs	Maintained / reactive. Fannie Mae service-life tables; +5 yrs from NRC Canada roof study and manufacturer warranty terms	MEDIUM
Roof cost growth	4.6%/yr	BLS Producer Price Index, nonresidential roofing contractors, 2007–2025	HIGH
Replacement cost	\$9.50/sq ft	Culbertson (N. Virginia) contractor price ranges	LOW
Reactive repair cost	3× planned	Industry estimates; no primary study found	LOW
Storm damage	0.15%/yr	Of roof cost. FEMA NRI expected annual building loss from hail, wind, tornado, hurricane and winter weather	MEDIUM
PaxSeal fee	\$16,400/yr	PAX program estimate for this roof size. Your proposal will state the actual fee	ESTIMATE
Leak consequences	1.30× warehouse	Scales reactive repair costs. Order supported by school-closure, hospital infection-control and insurer water-damage evidence; exact values are estimates	LOW

Estimates, not a quote. Projections use Culbertson (N. Virginia) market research and industry data, in nominal dollars; actual costs will differ. The PaxSeal fee is an estimate and your proposal will state the actual price. Net cost credits the roof life remaining at year 20.